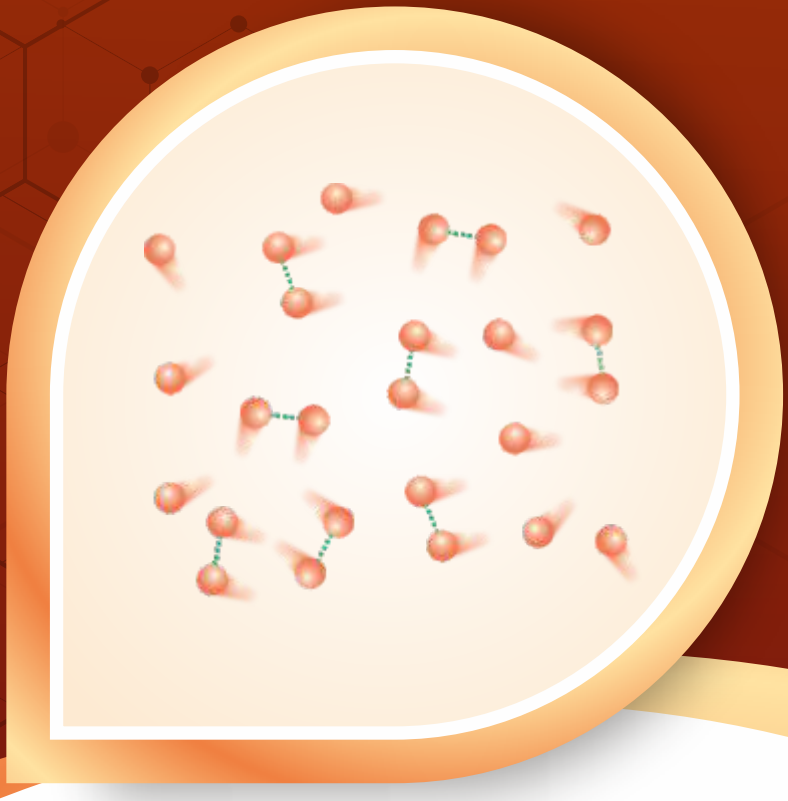
ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

path to success

PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Real Gas

(English Medium)

EXERCISE # S-I

1. Calculate the pressure (in atm) exerted by 22 g of carbon dioxide in 0.5 dm^3 at 300 K using:
(a) the ideal gas law and
(b) Van der Waal's equation respectively.
Given : [a = 3.6 atm litre^{2}mol^{-2} , b = 0.04 litre mol^{-1}, R = 0.08 L-atm/K-mol]

RG0001

2. Calculate from the Van der Waals equation, the temperature (in Kelvin) at which 192 g of SO_2 would occupy a volume of 6 dm^3 at 15 atm pressure.
[a = 5.68 atm L^2 mol^{-2}, b = 0.06 L mol^{-1}, R = 0.0821 L-atm/K-mol]

RG0002

3. The density of water vapour at 328.0 atm and 800 K is 135.0 g/dm^3. Determine (i) the molar volume (in L mol^{-1}) and (ii) the compression factor of water vapour . (R = 0.082 L-atm/K-mol)

RG0003

4. At 300 K and under a pressure of 10.1325 MPa, the compressibility factor of O_2 is 0.9. Calculate the mass (in kg) of O_2 necessary to fill a gas cylinder of 45 dm^3 capacity under the given conditions.
[R = 0.08L-atm/K-mol]

RG0004

5. 1 mole of CCl_4 vapours at 27°C occupies a volume of 40 L. If Van der Waals constants are 24.6 L^2 atm mol^{-2} and 0.5 $Lmol^{-1}$, then, calculate compressibility factor in
(a) Low pressure region (b) High Pressure region [R = 0.082 L -atm/K-mol]

RG0005

6. If at 200 K & 500 atm, density of CH_4 is 0.25 gm/ml then its compressibility factor (Z) is
(R = 0.08 L-atm/K-mol)

RG0006

7. Certain mass of a gas occupy 500 ml at 2 atm and 27ºC. Calculate the volume (in L) occupied by same mass of the gas at 0.3 atm and 227ºC. The compressibility factors of gas at the given condition are 0.8 and 0.9, respectively. **RG0007**

BOYLE TEMPERATURE

8. The vander waal's constant for a gas are a = 1.92 atm L^2 mol^{-2}, b = 0.06L mol^{-1}. If R = 0.08 L atm $K^{-1}mol^{-1}$, what is the Boyle's temperatrue (in K) of this gas.

RG0008

9. The Van der Waals constant for O_2 are a = 1.64 atm L^2 mol^{-2} and b = 0.04 L mol^{-1}. Calculate the temperature (in °C) at which O_2 gas behaves ideally for long range of pressure, in low pressure region.
(R = 0.082 L-atm/K-mol)

RG0009

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LIQUIFICATION OF GASES, CRITICAL PHENOMENON

10. The Van der Waals constants for gases A, B and C are as follows

Gas	a[/atm L^2 mol^{-2}]	b[/L mol^{-1}]
A	8.21	0.050
B	4.105	0.030
C	1.682	0.040

Which gas has (i) the highest critical temperature, (ii) the largest molecular volume, and (iii) most ideal behaviour around 500 K?

RG0010

11. For a real gas, if at critical conditions molar volume of gas is 8.2 litre at 3 atm, then critical temperature (in K) will be :

RG0011

12. An unknown gas behaves ideally at 540K in low pressure region, then calculate the temperature (in K) below which it can be liquified by applying pressure.

RG0012

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EXERCISE # S-II

1. The density of mercury is 13.6 g/cm^3. Estimate the value of **'b'** (in cm^3/ mole). (Hg = 200)

RG0013

2. The molarity of O_2 gas at 72 atm and 300K is 6M. Calculate the value of Z for O_2.

(Use : R = 0.08 atm-litre/K-mole).

RG0014

3. Calculate the amount of He (in gm) present in the 10 litre container at 240 atm and 300K. Given value of "b" for He is 0.08 dm^3 mol^{-1}. (R = 0.08 atm lit mol^{-1} K^{-1})

RG0015

4. For a real gas (mol. mass = 30) if density at critical point is 0.40 g/cm^3 and its $T_c = \frac{10^4}{41}$ K, then calculate Van der Waals constant a (in atm L^{2}mol^{-2}).

RG0016

5. Calculate the volume (in L) occupied by 0.2 mole of a Vander waal gas at 27ºC and 0.082 atm.

[a = 4.10 L^2 atm mol^{-2}, b = $\frac{1}{6}$ Lmol^{-1}]

RG0017

6. At what pressure (in atm) and 127ºC, the density of O_2 gas becomes 1.6 g/L ?

[a = 4.0 atm L^2 mol^{-2} , b = 0.4 Lmol^{-1}, R = 0.08 L - atm/ K-mol]

RG0018

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EXERCISE # O-I

1. The correct expression for the Van der Waals equation of state is :

(A) $\left(P+\frac{a}{n^2V^2}\right)(V-nb)=nRT$ (B) $\left(P+\frac{an^2}{V^2}\right)(V-nb)=\Delta nRT$

(C) $\left(P+\frac{an^2}{V^2}\right)(V-b)=nRT$ (D) $\left(P+\frac{an^2}{V^2}\right)(V-nb)=nRT$

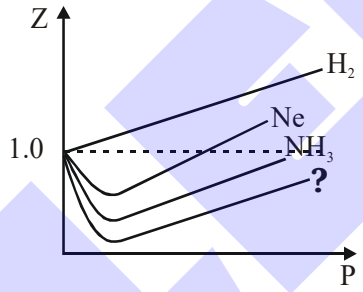
RG0019

2. At relatively high pressure, Van der Waals equation reduces to :

(A) $PV_m = RT$ (B) $PV_m = RT+\frac{a}{V_m}$ (C) $PV_m = RT + Pb$ (D) $PV_m = RT - \frac{a}{V_m^2}$

RG0020

3. Observe the following Z vs P graph.



The missing gas in the above graph can be :

(A) He (B) Ar (C) C_5H_{12} (D) All are correct

RG0021

4. Correct option regarding a container containing 1 mol of a gas in 22.4 litre container at 273 K is

(A) If compressibility factor (Z) > 1 then 'P' will be less than 1 atm.

(B) If compressibility factor (Z) > 1 then 'P' will be greater than 1 atm.

(C) If 'b' dominates, pressure will be less than 1 atm.

(D) If 'a' dominates, pressure will be greater than 1 atm.

RG0022

5. If 'V' is actual volume of 1 molecule of gas then, excluded volume (b) of 1 mole of gaseous molecule is -

(A) $4\ N_AV$ (B) N_AV (C) V/N_A (D) V

RG0023

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6. Consider the equation $Z = \frac{PV}{RT}$, Which of the following statements is correct :

(A) When Z > 1 real gases are easier to compress

(B) When Z = 1 real gases are easier to compress

(C) When Z > 1 real gases are difficult to compress

(D) When Z < 1 real gases are difficult to compress

RG0024

7. Compressibility factor of ideal gas is :-

(A) z > 1 (B) z > 1 (C) z = 1 (D) z = ∞

RG0025

8. The density of a gaseous substance at 1 atm pressure and 750 K is 0.30 g/lt. If the molecular weight of the substance is 27, the dominant forces existing among gas molecules is -

(A) Attractive (B) Repulsive (C) Both (A) and (B) (D) None of these

RG0026

9. The third virial coefficient of a He gas is 4×10^{-2} (lit/mol)^2. What will be volume of 2 mole He gas at 1 atm and 273K (273K > T_B)

(A) 22.0 L (B) 44.0 L (C) 44.8 L (D) 45.2 L

RG0027

10. At low pressure the vander waals equation is reduced to -

(A) $Z = \frac{pV_m}{RT} = 1 - \frac{a}{RTV_m}$ (B) $Z = \frac{pV_m}{RT} = 1 + \frac{a}{RT}p$

(C) $pV_m = RT$ (D) $Z = \frac{pV_m}{RT} = 1 - \frac{a}{RT}$

RG0028

11. The values of Van der Waals constant 'a' for the gases O_2, N_2, NH_3 and CH_4 are 1.360, 1.390, 4.170 and 2.253 L atm mol^{-2} respectively. The gas which can most easily be liquefied is :

(A) O_2 (B) N_2 (C) NH_3 (D) CH_4

RG0029

12. The values of critical temperatures of few gases are given gases :

Gases :	H_2	He	O_2	N_2
$\mathbf{T_C(K)}$	33.2	5.2	154.3	126

From the above data arrange the given gases in the increasing order of ease of their liquification.

(A) O_2, N_2, H_2, He (B) He, N_2, O_2, H_2

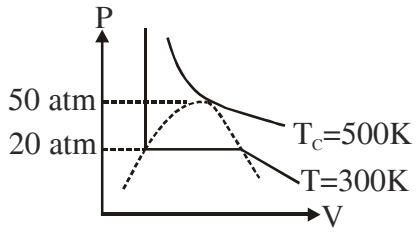
(C) He, H_2, N_2, O_2 (D) H_2, N_2, O_2, He

RG0030

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EXERCISE # O-II

1. For real gas the P–V curve was experiementally plotted and it had the following appearance. With respect to liquification, choose the incorrect statement :



(A) At T = 500 K, P = 40 atm, the state will be liquid

(B) At T = 300 K, P = 50 atm, the state will be gas

(C) At T < 300 K, P = 20 atm, the state will be gas

(D) At 300 K < T < 500 K, P > 50 atm, the state will be liquid

RG0031

2. Select the incorrect statement (s)

(A) The critical constant for a Vander Waal's gas is $V_C = 3b$, $P_C = \frac{a}{27b^2}$ and $T_C = \frac{a}{27Rb}$

(B) At 56 K a gas may be liquified if its critical temperature is –156°C.

(C) U_{avg} of gas in a rigid container can be doubled when the pressure is quadrupled by pumping in more gas at constant temperature

(D) At extremely low pressure, all real gases behave ideally.

RG0032

3. A 1 litre vessel contains 2 moles of a vanderwaal's gas.

Given data : $a = 2.5$ atm-L^2 mole^{-2} $T = 240$ K

$b = 0.4$ L-mole^{-1} $RT = 20$ L-atm mole^{-1}

Identify the correct options about the gas sample :

(A) Pressure of gas = 190 atm

(B) Compressibility factor = 4.75

(C) Attraction forces are dominant in the gaseous sample

(D) T_B (Boyle temperature) = 75 K

RG0033

4. Choose the correct statement(s) among the following -

(A) A gas having higher value T_C is easy to liquify

(B) The radius of molecules of gas having same value if T_C/P_C is same

(C) Hydrogen gas can be liquified at its boyle temperature by application of pressure.

(D) Real gas show negative deviation from ideal behaviour at low pressure condition.

RG0034

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5. Select the **INCORRECT** statement(s):

(A) At Boyle's temperature, a real gas behaves like an ideal gas irrespective of pressure.

(B) At critical condition, a real gas behaves like an ideal gas.

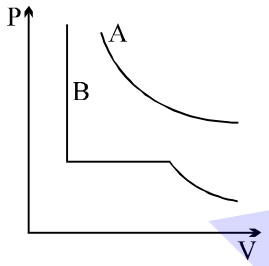
(C) At low pressure, Van der Waal's constant 'a' dominates over 'b'.

(D) At high pressure, Van der Waal's constant 'b' dominates over 'a'.

RG0050

Question No. 6 & 7 (2 questions)

For two gases A and B, P v/s V isotherms are shown at same temperature, T K . T_A & T_B are critical temperatures of A & B respectively



6. Which of the following is true?

(A) $T_A < T < T_B$ (B) $T_A > T > T_B$

(C) $T_A > T_B > T$ (D) none of above

RG0036

7. The correct statement(s) is/are

(I) Pressure correction term will be more negligible for gas B at T K.

(II) The curve for gas 'B' will be of same shape as for gas A if $T > T_B$

(III) Gas 'A' will show same P v/s V curve as of gas 'B' if $T > T_A$

(A) III only (B) II and III (C) II only (D) All

RG0037

Match the column:

8. Match the column :

	Column-I		**Column-II**
(A)	Boyle's temperature	(P)	Depends on 'a' and 'b'
(B)	Compressibility factor	(Q)	Depends on identity of real gas
(C)	Real gas with very large molar volume	(R)	The temperature at which $\frac{dZ}{dP} = 0$ at low pressure region.
(D)	Critical temperature	(S)	PV = nRT
		(T)	$\frac{8a}{27Rb}$

RG0038

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EXERCISE # JEE-MAINS

1. 'a' and 'b' are Van der Waals constants for gases. Chlorine is more easily liquefied than ethane because :- **[AIEEE-2011]**

(1) a for Cl_2 < a for C_2H_6 but b for Cl_2 > b for C_2H_6

(2) a for Cl_2 > a for C_2H_6 but b for Cl_2 < b for C_2H_6

(3) a and b for Cl_2 > a and b for C_2H_6

(4) a and b for Cl_2 < a and b for C_2H_6

RG0039

2. If Z is the compressibility factor, Van der Waals equation **at low** pressure can be written as : **[JEE-MAINS-2014]**

(1) $Z = 1 - \frac{Pb}{RT}$ (2) $Z = 1 + \frac{Pb}{RT}$ (3) $Z = 1 + \frac{RT}{Pb}$ (4) $Z = 1 - \frac{a}{V_m RT}$

RG0041

3. When does a gas deviate the most from it's ideal behaviour ? **[JEE-MAINS(ONLINE)-2015]**

(1) At high pressure and low temperature

(2) At high pressure and high temperature

(3) At low pressure and low temperature

(4) At low pressure and high temperature

RG0040

4. Among the following, the incorrect statement is : **[JEE-Mains-2017(ONLINE)]**

(1) At low pressure, real gases show ideal behaviour

(2) At very large volume, real gases show ideal behaviour

(3) At Boyle's temperature, real gases show ideal behaviour

(4) At very low temperature, real gases show ideal behaviour

RG0042

5. The volume of gas A is twice than that of gas B. The compressibility factor of gas A is thrice than that of gas B at same temperature. The pressures of the gases for equal number of moles are : **[JEE-MAINS-2019]**

(1) $2P_A = 3P_B$ (2) $P_A = 3P_B$ (3) $P_A = 2P_B$ (4) $3P_A = 2P_B$

RG0043

6. At a given temperature T, gases Ne, Ar, Xe and Kr are found to deviate from ideal gas behaviour.

Their equation of state is given as $p = \frac{RT}{V - b}$ at T. **[JEE-MAINS-2019]**

Here, b is the van der Waals constant. Which gas will exhibit steepest increase in the plot of Z (compression factor) vs p?

(1) Ne (2) Ar (3) Xe (4) Kr

RG0044

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7. Consider the van der Waals constants, a and b, for the following gases.

Gas	Ar	Ne	Kr	Xe
a/ (atm dm^6 mol^{-2})	1.3	0.2	5.1	4.1
b/ (10^{-2} dm^3 mol^{-1}	3.2	1.7	1.0	5.0

[JEE-MAINS-2019]

Which gas is expected to have the highest critical temperature?

(1) Kr (2) Ne (3) Ar (4) Xe

RG0045

8. Consider the following table : **[JEE-MAINS-2019]**

Gas	a/(k Pa dm^6 mol^{-1})	b/(dm^3 mol^{-1})
A	642.32	0.05196
B	155.21	0.04136
C	431.91	0.05196
D	155.21	0.4382

a and b are vander waals constant. The correct statement about the gases is :

(1) Gas C will occupy lesser volume than gas A; gas B will be lesser compressible than gas D

(2) Gas C will occupy more volume than gas A; gas B will be lesser compressible than gas D

(3) Gas C will occupy more volume than gas A; gas B will be more compressible than gas D

(4) Gas C will occupy lesser volume than gas A; gas B will be more compressible than gas D

RG0046

9. A certain gas obeys $P(V_m - b) = RT$. The value of $\left(\frac{\partial Z}{\partial P}\right)_T$ is $\frac{xb}{RT}$. The value of x is_____.

(Integer answer) (Z : compressibility factor) **[JEE-MAINS-2021]**

RG0051

10. The unit of the van der Waals gas equation parameter 'a' in $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$ is :

(1) kg m s^{-2} (2) dm^3 mol^{-1} **[JEE-MAINS-2021]**

(3) kg m s^{-1} (4) atm dm^6 mol^{-2}

RG0052

RG0052

11. For a real gas at 25°C temperature and high pressure (99 bar) the value of compressibility factor is 2, so the value of Vander Waal's constant 'b' should be_____ × 10^{-2} L mol^{-1} (Nearest integer)

(Given R = 0.083 L bar K^{-1} mol^{-1}) **[JEE-MAINS-2022]**

RG0053

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EXERCISE # JEE-ADVANCED

1. The compression factor (compressibility factor) for one mole of a Van der Waals' gas at 0° C and 100 atmosphere pressure is found to be 0.5. Assuming that the volume of a gas molecule is negligible, calculate the Van der Waals' constant 'a'. **[JEE 2001]**

RG0054

2. The density of the vapour of a substance at 1 atm pressure and 500 K is 0.36 Kg m^{-3}. The vapour effuses through a small hole at a rate of 1.33 times faster than oxygen under the same condition. Determine **[JEE 2002]**

(i) mol. wt. ;

(ii) molar volume ;

(iii) compression factor z of the vapour and

(iv) which forces among the gas molecules are dominating, the attractive or the repulsive

RG0055

3. Positive deviation from ideal behaviour takes place because of **[JEE 2003]**

(A) molecular attractions between atoms and $\frac{PV}{nRT} > 1$

(B) molecular attractions between atoms and $\frac{PV}{nRT} < 1$

(C) finite size of atoms and $\frac{PV}{nRT} > 1$

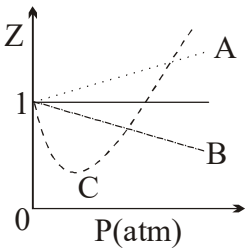
(D) finite size of atoms and $\frac{PV}{nRT} < 1$

RG0056

4. For a real gas obeying van der Waals' equation a graph is plotted between PV_m (y-axis) and P(x-axis) where V_m is molar volume. Find y-intercept of the graph. **[JEE 2004]**

RG0057

5. The given graph represents the variation of Z (compressibility factor = $\frac{PV}{nRT}$) versus P, for three real gases A, B and C. Identify the only **INCORRECT** statement.



(A) for the gas A, a = 0 and its dependence on P is linear at all pressure

(B) for the gas B, b = 0 and its dependence on P is linear at all pressure

(C) for the gas C, which is typical real gas for which neither a nor b = 0. By knowing the minima and the point of intersection, with Z = 1, a and b can be calculated.

(D) At high pressure, the slope is positive for all real gases A, B and C. **[JEE 2006]**

RG0058

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6. Match gases under specific conditions listed in Column I with their properties / laws in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS.

Column I	**Column II**
(A) Hydrogen gas (P = 200 atm, T = 273 K)	(P) Compressibility factor ≠ 1
(B) Hydrogen gas (P ~ 0, T = 273 K)	(Q) Attractive forces are dominant
(C) CO_2 (P = 1 atm, T = 273 K)	(R) PV = nRT
(D) Real gas with very large molar volume	(S) P (V– nb) = nRT

[JEE 2007]

RG0059

7. A gas described by van der Waals' equation **[JEE 2008]**

(A) behaves similar to an ideal gas in the limit of large molar volumes

(B) behaves similar to an ideal gas in the limit of large pressures

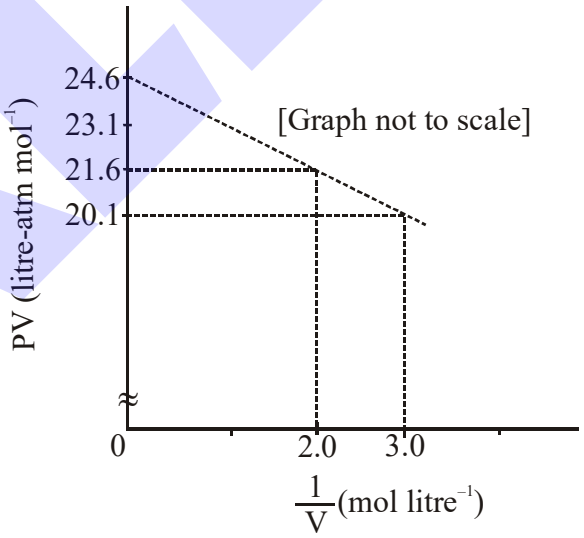
(C) is characterised by van der Waals' coefficients that are dependent on the identity of the gas but are independent of the temperature

(D) has the pressure that is lower than the pressure exerted by the same gas behaving ideally

RG0060

8. For one mole of a Van der Waals gas when b = 0 and T = 300 K, the PV vs. 1/V plot is shown below.

The value of the Van der Waals constant a (atm. litre^2 mol^{-2}) is **[JEE 2012]**



(A) 1.0 (B) 4.5 (C) 1.5 (D) 3.0

RG0047

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ANSWER KEY

EXERCISE # S-I

1. (a) 24.00, (b) 21.40 **2. 388.00** **3. (i) 0.13, (ii) 0.66 or 0.67**

4. 6.66 or 6.67 **5. (a) 0.98, (b) 1.01** **6. 2.00**

7. 6.25 **8. 400.00** **9. 227.00**

10. (i) A, (ii) A, (iii) C **11. 800.00** **12. 160.00**

EXERCISE # S-II

Que.	1	2	3	4	5	6
Ans.	58.82	0.50	222.22	1.68 or 1.69	60.00	1.62

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	C	C	B	A	C	C	B	D	A
Que.	11	12								
Ans.	C	C								

EXERCISE # O-II

Que.	1	2	3	4	5	6	7
Ans.	A,B,C	A,C	A,B,D	A,B,D	A,B	A	C
Que.	11	12	13	14			
Ans.	A-P,Q,R,S	B-P,Q	C-S	D-P,Q,T			

EXERCISE # JEE-MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	4	1	4	1	3	1	3	1	4
Que.	11									
Ans.	25									

EXERCISE # JEE-ADVANCED

1. **1.256 atm L^2 mol^{-2}**

2. **(i) 18 g/mol,** **(ii) 50 L mol^{-1},** **(iii) 1.218 ,** **(iv) repulsive**

3. **(C)** **4. RT** **5. (D)**

6. (A) - P, S ; (B) - R ; (C) - P, Q ; (D) - R **7.** **(A,C,D)** **8. (C)**

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